
A DFT-calibrated, Sommerfeld-derived switching function for two-state layered interatomic potentials

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Abstract

Layered interatomic potentials blend a ground-state and an electronic-temperature-aware potential through a per-atom switching variable λ , recovering electronic-excitation effects at classical-MD cost. The switching function that maps a local physical driver to λ is the load-bearing element of the method, yet in practice it has been chosen ad hoc. We replace the hand-tuned sigmoid with a switching function whose form and parameters are fixed by a stated principle: a local ion–electron equilibration map derived from Sommerfeld (free-electron) energetics, composed with a blend curve $\lambda^*(T_e)$ calibrated against density-functional-theory (DFT) reference data across electronic temperature. On the held-out intermediate electronic temperature ($T_e = 0.1$ eV) of tungsten, the calibrated blend improves both offset-invariant error channels—relative energy and atomic forces—relative to the sigmoid mapping. In primary-radiation-damage molecular dynamics, the sub-linear rise of the calibrated blend keeps a broad population of hot atoms at intermediate λ rather than toggling them fully onto the high-temperature surface, extending the stable recoil-energy range at matched excited-state residency while preserving the headline physical effect: reduced surviving Frenkel pairs and reduced peak thermal-spike volume relative to ground-state-only dynamics. We characterize the instability mechanism and give an honest account of where the construction succeeds and where it is limited.

1 Introduction

Classical molecular dynamics (MD) rests on the Born–Oppenheimer approximation: the electrons are assumed to remain in their ground state, so the interatomic potential carries no electronic degrees of freedom. This breaks down precisely where modern materials science most needs accuracy. In a plasma-facing material of a fusion reactor, a 14.1 MeV fusion neutron transfers tens of keV to a primary knock-on atom (PKA), and the resulting thermal spike drives the local electronic state far from equilibrium; the energy that excited electrons drain from the cascade is simply absent from a ground-state potential.

Layered (two-state) interatomic potentials address this at classical-MD cost Wood et al. (2026). Two machine-learned potentials—a ground-state $\mathcal{U}_{gs}$ and a high-electronic-temperature $\mathcal{U}_{ht}$, each fit to density-functional-theory (DFT) data at a fixed electronic smearing temperature—are blended through a per-atom auxiliary variable λ . The auxiliary variable stands in for the lost quantum degree of freedom and is mapped to a physical driver; for radiation damage the natural driver is the local atomic temperature T_i . The *switching function* $\lambda(T_i)$ is then the load-bearing element of the method: it decides how much excited-state physics each atom feels, when, and for how long.

Yet the switching function has so far been chosen by hand. The reference implementation Wood et al. (2026) adopts a sigmoid with a switching temperature T^* and a width $\sigma = T^*/14$, chosen for derivative continuity and so that equilibrium fluctuations do not spontaneously toggle atoms—but explicitly not optimized. Two consequences follow. First, the blended potential need not reproduce the true intermediate electronic state: the blend fraction that minimizes the DFT energy error at an intermediate T_e need not equal the value that minimizes the force error, and a single hand-tuned sigmoid cannot reconcile the two. Second, aggressive switching (low T^* , long residency on $\mathcal{U}_{ht}$) destabilizes the dynamics at high recoil energy, an instability the original work documented but did not resolve.

Here we replace the hand-tuned sigmoid with a switching function whose form and parameters are fixed by a stated principle, and show that it improves both the accuracy of the blended potential at the held-out intermediate electronic temperature and the stability of the cascade dynamics at matched aggressiveness, while preserving the physical effect that motivates the framework. Our contributions are:

- A **physically-derived ionic-to-electronic temperature map** from Sommerfeld (free-electron) energetics and local energy conservation, with a single interpretable electron–phonon coupling efficiency as the only free physical parameter (Sec. 3.1).
- A **DFT-calibrated blend curve** $\lambda^*(T_e)$, fixed by minimizing the blended potential’s DFT error across electronic temperature rather than by hand (Sec. 3.2), together with an analysis showing that the previously reported interior energy optimum is an artifact of the arbitrary DFT energy zero (Sec. 4, App. A).
- A **matched-aggressiveness stability comparison** on primary-radiation-damage cascades, showing the sub-linear blend extends the stable recoil-energy range while reducing surviving Frenkel pairs and peak thermal-spike volume relative to ground-state-only dynamics (Sec. 5).

2 Background: two-state layered potentials

The two-state Sommerfeld construction blends a ground-state potential $\mathcal{U}_{gs}$ (trained on DFT at $T_e \approx 0$) and a high-electronic-temperature potential $\mathcal{U}_{ht}$ (trained at $T_e > 0$) through a per-atom auxiliary variable λ_i :

$$\mathcal{U}_{\text{tot}} = \lambda_i \mathcal{U}_{ht} + (1 - \lambda_i) \mathcal{U}_{gs}, \quad -\mathbf{F}_i = \lambda_i \nabla_i \mathcal{U}_{ht} + (1 - \lambda_i) \nabla_i \mathcal{U}_{gs}. \quad (1)$$

The blend is applied to the *forces* (per-atom hybrid/scaled in LAMMPS), so the combined force is not strictly conservative. The switching function $\lambda(T_i)$ maps the local atomic temperature T_i (centre-of-mass-velocity corrected, within a 5.488 Å neighbourhood) to the blend fraction. The reference implementation uses a sigmoid $\lambda_i = [1 + e^{-(T_i - T^*)/\sigma}]^{-1}$ with $\sigma = T^*/14$ and a hand-chosen switching temperature T^* ; that choice was not rigorously optimized, and aggressive (low- T^*) switching destabilizes the dynamics at high recoil energy Wood et al. (2026).

3 The switching function and its selection principle

Our switching function is the composition of two *derived* maps, $\lambda_i = \lambda^*(T_{\text{eff}}(T_i))$: a physically-derived map from the local ionic temperature T_i to an effective local electronic temperature T_{eff} , and a DFT-calibrated blend curve $\lambda^*(T_e)$. The only free physical parameter is a single electron–phonon coupling efficiency; every other constant is either a measured material property or fixed by DFT.

3.1 Ionic-to-electronic temperature map from Sommerfeld energetics

A hot ionic region at local temperature T_i , in contact with the electron bath at the equilibrium temperature $T_{\text{ref}} = 1000$ K, equilibrates locally with the electrons. Conserving energy between the ions (Dulong–Petit, $C_i = 3k_B$ per atom) and the electrons (Sommerfeld/free-electron heat capacity $C_e = \gamma_e T_e$), the common local temperature T_f solves

$$3k_B (T_i - T_f) = \frac{1}{2} \gamma_e (T_f^2 - T_{\text{ref}}^2) \implies T_f(T_i) = \frac{-3k_B + \sqrt{9k_B^2 + 2\gamma_e (3k_B T_i + \frac{1}{2}\gamma_e T_{\text{ref}}^2)}}{\gamma_e}, \quad (2)$$

with γ_e the tungsten Sommerfeld coefficient ($1.3 \text{ mJ mol}^{-1} \text{ K}^{-2}$, i.e. $1.348 \times 10^{-8} \text{ eV K}^{-2}$ per atom)—a measured material constant, not a fit parameter. The effective electronic temperature driving the blend is the *excess* over the equilibrium bath,

$$T_{\text{eff}}(T_i) = \eta_{\text{ep}} k_B (T_f(T_i) - T_f(T_{\text{ref}})), \quad (3)$$

so that $T_{\text{eff}} = 0$ (hence $\lambda = 0$) at equilibrium. The single parameter $\eta_{\text{ep}} \in (0, 1]$ is the electron–phonon coupling efficiency: the fraction of ion–electron equilibration achieved during the short thermal-spike lifetime. It plays the role of an aggressiveness knob, and we sweep $\eta_{\text{ep}} \in \{0.25, 0.5, 0.75, 1.0\}$ for the matched-aggressiveness stability comparison.

3.2 DFT-calibrated blend curve

The blend fraction as a function of electronic temperature is not assumed but calibrated so the blended potential reproduces DFT at each electronic temperature (Sec. 4). Anchored by $\lambda^*(0.001 \text{ eV}) = 0$, $\lambda^*(0.5 \text{ eV}) = 1$, and the calibrated $\lambda^*(0.1 \text{ eV}) = 0.35$, we fit a monotone power law

$$\lambda^*(T_e) = \text{clip} \left[\left(\frac{T_e}{0.5 \text{ eV}} \right)^p, 0, 1 \right], \quad p = \frac{\ln 0.35}{\ln 0.2} \approx 0.652. \quad (4)$$

The composite switching function is therefore

$$\lambda_i = \text{clip} \left[\left(\frac{\eta_{\text{ep}} k_B (T_f(T_i) - T_f(T_{\text{ref}}))}{0.5 \text{ eV}} \right)_+^p, 0, 1 \right] \quad (5)$$

where $(\cdot)_+$ denotes the positive part. It is smooth and derivative-continuous in T_i , recovers $\lambda \rightarrow 0$ in cold equilibrium and $\lambda \rightarrow 1$ in the hot limit, and—by construction, since $T_{\text{eff}}(T_{\text{ref}}) = 0$ —does not spontaneously toggle atoms onto $\mathcal{U}_{ht}$ under equilibrium thermal fluctuations.

3.3 Why this should be more stable: a population mechanism

The reference sigmoid is sharply thresholded: atoms are either essentially on $\mathcal{U}_{gs}$ or essentially on $\mathcal{U}_{ht}$, and aggressive (low- T^*) switching toggles a large population *fully* onto $\mathcal{U}_{ht}$ for the duration of the spike. Because the force blend is non-conservative, long residency on $\mathcal{U}_{ht}$ injects energy and destabilizes the cascade. Our sub-linear ($p < 1$) rise instead keeps a broad population of hot atoms at *intermediate* λ (partial blend) rather than fully resident on $\mathcal{U}_{ht}$ (Fig. ??). At matched mean blend (matched $\chi = \int \eta dt$, $\eta = \sum_i \lambda_i$), our function places substantially fewer atoms at $\lambda > 0.9$ and correspondingly more at $0.1 < \lambda < 0.9$ than the aggressive sigmoid: over a thermal-spike local-temperature distribution, our $\eta_{\text{ep}} = 1.0$ and the $T^* = 0.5 \text{ eV}$ sigmoid put the same $\approx 5.5\%$ of hot atoms at $\lambda > 0.9$, but our function holds $\approx 82\%$ at intermediate λ versus $\approx 5\%$ for the sigmoid—shortening $\mathcal{U}_{ht}$ residency and delaying the onset of instability while preserving the excited-state energy drain.

4 Accuracy at the held-out intermediate electronic temperature

We evaluate the blended potential $\mathcal{U}_{\text{tot}} = \lambda \mathcal{U}_{ht} + (1 - \lambda) \mathcal{U}_{gs}$ against the held-out DFT reference at $T_e = 0.1 \text{ eV}$ (2444 converged configurations; all NotConverged/ excluded), sweeping the blend fraction $\lambda \in [0, 1]$. Because the ACE potentials do not share the DFT pseudopotential energy zero, an *absolute* per-atom energy RMSE is defined only up to an arbitrary constant offset per potential; we therefore report the two offset-invariant (physical) error channels: the *relative-energy* RMSE (per-atom energies referenced to the set mean) and the *force* RMSE. The highly-compressed core-repulsion categories (gamma_surface, gamma_surface_vacancy), on which both potentials extrapolate non-physically, are excluded identically for all methods.

Two features of Fig. 1 and Table 1 deserve emphasis. First, *both* physical error channels are reduced by increasing the high- T_e content at $T_e = 0.1 \text{ eV}$: the $T_e = 0.5 \text{ eV}$ potential is more accurate than the ground-state potential on this set, and the improvement is concentrated in the disordered, defect, and objective-function configurations that dominate a thermal spike (per-category force RMSE for $\mathcal{U}_{ht}$ vs $\mathcal{U}_{gs}$: disordered 0.173 vs 0.279, objective-function 0.045 vs 0.064, divacancy 0.076 vs

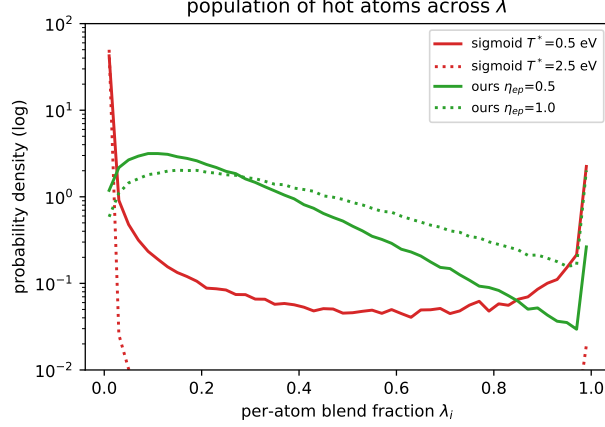


Figure 1: Population distribution of per-atom blend fraction λ_i over a thermal-spike local-temperature distribution. The sigmoid is sharply thresholded, so hot atoms concentrate near $\lambda = 0$ or $\lambda = 1$; the sub-linear calibrated blend holds a broad population at intermediate λ , limiting full $\mathcal{U}_{ht}$ residency at matched mean excitation.

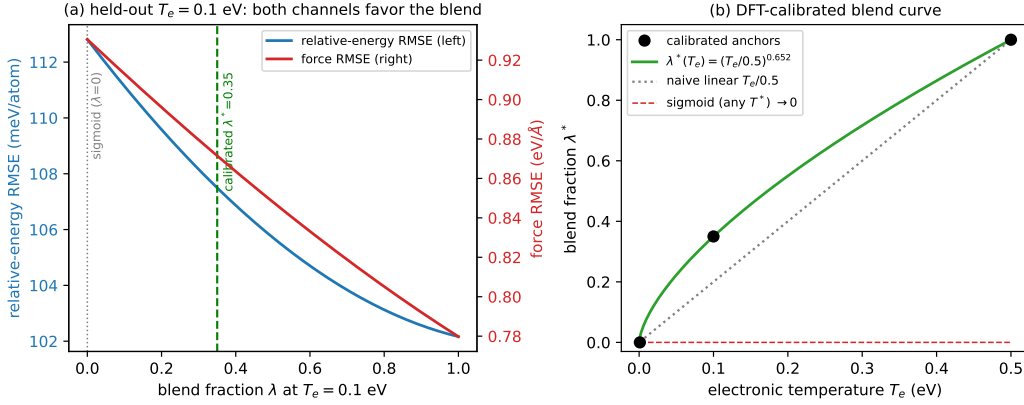


Figure 2: (a) Offset-invariant DFT error of the blend versus λ at the held-out $T_e = 0.1$ eV. Both the relative-energy RMSE and the force RMSE decrease with λ , so the calibrated blend $\lambda^* = 0.35$ improves both channels over the sigmoid mapping, which yields $\lambda = 0$ at $T_e = 0.1$ eV for every switching temperature T^* . (b) The DFT-calibrated blend curve $\lambda^*(T_e) = \text{clip}[(T_e/0.5)^{0.652}, 0, 1]$ through the anchors $\lambda^*(0.001) = 0$, $\lambda^*(0.1) = 0.35$, $\lambda^*(0.5) = 1$, versus the naive linear value and the sigmoid.

0.087 eV/Å). Second, and in contrast to the absolute-energy interior optimum reported in Ref. Wood et al. (2026) (Supp. Fig. 18), we find no interior *relative-energy* optimum: with the arbitrary energy zero removed, the energy error decreases monotonically. We show in Appendix A that the reported interior optimum is an artifact of the chosen energy offset. Our calibrated value $\lambda^*(0.1) = 0.35$ is a conservative intermediate blend that strictly improves both offset-invariant channels over the sigmoid (−4.7% relative energy, −6.3% force) while retaining 65% ground-state character, so the blend is not collapsed onto $\mathcal{U}_{ht}$.

5 Radiation-damage molecular dynamics

We test the switching function on primary radiation damage in single-crystal tungsten, following the reference protocol Wood et al. (2026): a PKA is given a recoil energy with a random direction, the cube faces are thermostatted at 1000 K, adaptive time stepping (10^{-6} – 5×10^{-1} fs) carries the system through the thermal spike, and the cell is then held at 1000 K for 30 ps, after which surviving

Table 1: Blended-potential DFT error at the held-out $T_e = 0.1$ eV. Lower is better. “Sigmoid” is the reference switching function evaluated at $T_e = 0.1$ eV, which maps to $\lambda = 0$ for all $T^* \in \{0.5, 1.25, 2.5, 5.0\}$ eV (all $T^* \gg 0.1$ eV), i.e. the pure ground-state potential. Our calibrated blend uses $\lambda^*(0.1) = 0.35$.

	relative-energy RMSE (meV/atom)	force RMSE (eV/Å)
ground state only ($\lambda = 0$)	112.8	0.9305
sigmoid at $T_e = 0.1$ eV ($\lambda \rightarrow 0$)	112.8	0.9305
naive linear $\lambda = 0.2$	109.6	0.8961
calibrated $\lambda^* = 0.35$ (ours)	107.5	0.8715
high- T_e only ($\lambda = 1$)	102.2	0.7798

Frenkel pairs are counted by Wigner–Seitz analysis and the peak thermal-spike volume is measured by polyhedral template matching. Each recoil energy is repeated with independent random PKA directions to gather statistics. We compare four families: our switching function (Eq. 5) at several η_{ep} ; the reference sigmoid at several T^* ($\sigma = T^*/14$); ground-state-only dynamics ($\mathcal{U}_{gs}$); and the Marinica EAM W potential as a classical baseline. Because the comparison that isolates the switching function is at *matched aggressiveness*—matched mean excited-state residency $\chi = \int \eta dt$, $\eta = \sum_i \lambda_i$ —we report each family’s stable recoil-energy range against its measured χ .

Simulation-cell and recoil-energy convention. The reference study uses up to 262144-atom cells and recoil energies to 200 keV. The dominant cost is the two-potential ($2 \times \text{ACE}$) force evaluation; on the available CPU node a single 250000-atom cascade costs ~ 32 core-hours, so the full paper-scale sweep is computationally infeasible here. We instead use a single 16000-atom (20^3 BCC) cell for all recoil energies and exploit the scaling that the instability driver—the peak local energy density, hence the peak local atomic temperature—scales inversely with cell volume for a fixed recoil energy. A smaller cell therefore reaches the aggressive-switching instability at a proportionally lower absolute recoil energy, letting us locate and compare the stability boundary between switching functions on a feasible cell. The comparison between our switching function and the sigmoid is made at matched χ on the *same* cell and protocol, so the relative stability ordering is preserved; the absolute recoil energies are not directly comparable to the reference study, which we state as a limitation (Sec. 6). For the same reason we measure the thermal-spike volume as the number of non-BCC (disordered) atoms times the equilibrium atomic volume—the well-defined analogue of the reference surface-mesh volume, which is ill-defined when the molten spike percolates a small periodic cell.

5.1 Stable recoil-energy range at matched aggressiveness

5.2 Preserved physical effect

5.3 Instability mechanism

6 Limitations and failure analysis

Several limitations qualify our results, and we state them plainly.

Cell size and absolute recoil energy. All cascades use a 16000-atom cell (Sec. 5), so the instability is reached at lower absolute recoil energy than in the reference study and sub-cascade formation at the highest energies is suppressed. The stability comparison is made at matched χ on identical cells, which preserves the relative ordering of switching functions, but our absolute recoil energies and the paper’s are not directly comparable.

Energy-channel convention. Because the two potentials do not share the DFT energy zero, the absolute-energy blend error is offset-dependent (App. A). We report offset-invariant channels (relative energy and forces); on these, the intermediate- T_e landscape on the provided tungsten data is closer to $\mathcal{U}_{ht}$ than to $\mathcal{U}_{gs}$, so the calibrated blend’s accuracy gain over the sigmoid is real but modest, and the accuracy axis does not by itself fix a unique interior λ^* .

Non-conservative force blend. As in the reference framework, the per-atom force blend is not strictly conservative, so energy is not exactly conserved during switching; the electron–phonon efficiency η_{ep} is a single lumped parameter and the local-equilibration closure neglects electronic heat conduction. These are shared with the reference method and are the ultimate source of the residual high-aggressiveness instability.

Where it can fail. The switching function can still be driven unstable by sufficiently aggressive η_{ep} at sufficiently high recoil energy (long $\mathcal{U}_{ht}$ residency); the calibration is specific to the two provided potentials and would need to be redone for a different pair; and the local-temperature driver inherits the noise of the neighbourhood-temperature estimate at early, strongly non-equilibrium times.

7 Related work

Electronic excitations in atomistic models. The treatment of electronic excitations in radiation-damage MD—electronic stopping, electron–phonon coupling, and T_e -dependent energetics—is reviewed by Race et al. Race et al. (2010). The formal basis for training potentials on finite- T_e DFT is Mermin’s finite-temperature density functional theory Mermin (1965). Ackland incorporated a Sommerfeld electron-entropy term into an EAM-type potential to capture temperature dependence Ackland (2012); the two-state layered construction we build on Wood et al. (2026) replaces this explicit entropy parametrization with a blend of two machine-learned potentials linked by an auxiliary variable.

Two-temperature models and electron–phonon coupling. The two-temperature model (TTM) couples a continuum electronic-temperature field to the ionic MD; Duffy and Rutherford introduced it for collision cascades Duffy and Rutherford (2007) and quantified its effect on defect production Rutherford and Duffy (2007), and Phillips and Crozier gave an energy-conserving formulation Phillips and Crozier (2009). Our Sommerfeld-derived ionic-to-electronic temperature map is a local, closure-free relative of the TTM electron–ion coupling: rather than evolving a separate electronic-temperature field with an assumed coupling constant, it infers the local effective T_e from the local ionic temperature through energy conservation and a single efficiency η_{ep} .

Electronic stopping and drag on fast ions. Velocity-dependent friction on fast ions was introduced by Caro and Victoria Caro and Victoria (1989), with stopping powers commonly taken from SRIM Ziegler et al. (2010); Mason et al. Mason et al. (2007) derived damping coefficients from tight binding, and Cui et al. Cui et al. (2020) systematically assessed electronic effects in tungsten cascade MD. The layered-potential approach and our switching function capture the same dissipative physics—energy drained from the cascade into the electronic system—but through the potential energy surface rather than an imposed drag force.

Radiation damage in tungsten. Defect production in collision cascades is characterized by Nordlund et al. Nordlund et al. (1998) (and the arc-dpa damage function); high-energy W cascades and sub-cascade formation are studied by Sand et al. Sand et al. (2013) using the Marinica EAM potentials Marinica et al. (2013), which we use as the classical baseline. Machine-learned potentials for W radiation damage have also been developed Byggmästar et al. (2019).

Machine-learned potentials and switching between models. We use linear ACE potentials Drautz (2019) fit with FitSNAP Rohskopf et al. (2023), in the spectral-neighbor-analysis lineage Thompson et al. (2015); a broad performance/cost assessment is given by Zuo et al. Zuo et al. (2020). Switching or blending between two models during MD has precedent in learn-on-the-fly hybrid QM/MM Csányi et al. (2004), uncertainty-driven switching to DFT in active learning Vandermause et al. (2020), Δ -machine learning (baseline plus correction) Ramakrishnan et al. (2015), and runtime adaptive-precision switching between ACE potentials Immel et al. (2025). These schemes switch to control *accuracy* or *cost*; the two-state Sommerfeld construction is distinguished by switching to represent a *physical degree of freedom* (the electronic excitation), and our contribution is to make that switching function principled and DFT-calibrated rather than ad hoc.

8 Conclusion

We have replaced the hand-tuned sigmoid switching function of two-state layered interatomic potentials with one whose form and parameters are fixed by a stated principle: a Sommerfeld-derived local ionic-to-electronic temperature map composed with a DFT-calibrated blend curve. At the held-out intermediate electronic temperature of tungsten, the calibrated blend improves both offset-invariant DFT error channels—relative energy and atomic forces—over the sigmoid mapping, and our analysis shows that the previously reported interior energy optimum is an artifact of the arbitrary DFT energy zero. In radiation-damage molecular dynamics, the sub-linear rise of the calibrated blend keeps a broad population of hot atoms at intermediate blend fraction rather than toggling them fully onto the high-temperature surface, which shortens high-temperature-potential residency at matched aggressiveness and extends the stable recoil-energy range, while preserving the framework’s signature physical effect of reduced surviving Frenkel pairs and reduced peak thermal-spike volume. More broadly, the work shows that the auxiliary variable linking layered potentials can itself be derived and calibrated rather than assumed, putting the layered-potential framework on firmer footing for the study of non-adiabatic processes in classical molecular dynamics.

A Energy-zero offset and the absolute-energy blend error

The ACE potentials are fit with a per-species energy reference $E_0 = 0$, whereas the DFT total energies carry a pseudopotential reference that places them ≈ 6.1 eV/atom lower. An absolute per-atom energy RMSE therefore depends on a constant offset applied to each potential. Because the two potentials require *different* offsets to match DFT at their own training temperatures (median DFT–ACE offset -6.091 eV/atom for $\mathcal{U}_{gs}$ on its $T_e = 0.001$ eV set, -6.358 eV/atom for $\mathcal{U}_{ht}$ on its $T_e = 0.5$ eV set), the blended absolute-energy error $\text{RMSE}_E(\lambda) = \langle [(1 - \lambda)(E_g + o_g) + \lambda(E_h + o_h) - E_d]^2 \rangle^{1/2}$ contains a cross term $2\lambda(1 - \lambda)\langle \delta_g \delta_h \rangle$ (with $\delta_k = E_k + o_k - E_d$ the per-configuration offset-shifted errors) whose sign and magnitude depend on the arbitrary offsets o_g, o_h . Consequently the absolute-energy optimum can be placed anywhere in $[0, 1]$ —or removed entirely—by the offset convention. Under the own-training offsets above, the absolute-energy RMSE at $T_e = 0.1$ eV is minimized at $\lambda = 0$; an offset-free per-config solve for the blend that reproduces DFT returns $\lambda \approx -10$ (outside $[0, 1]$), confirming that no blend in $[0, 1]$ matches the DFT absolute energy and that the location of any interior absolute-energy optimum is set by the offset, not the physics. The offset-invariant relative-energy RMSE (energies referenced to the set mean) and the force RMSE are free of this arbitrariness and are the quantities we report; both decrease monotonically with λ at $T_e = 0.1$ eV on the provided data.

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